

Probability Distributions

General Lecture Notes for Task List 03

Discrete models, random variables, assumptions, and interpretation

Prepared as conceptual lecture notes, not as solutions to the task list.

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Contents

1	Purpose of These Notes	3
2	From Experiments to Random Variables	3
2.1	Sample space and elementary outcomes	3
2.2	Random variables	4
2.3	Support of a random variable	4
2.4	Probability distribution	4
2.5	Distribution as compression of a sample space	5
3	Bernoulli Trials and the Bernoulli Distribution	5
3.1	The two-outcome trial	5
3.2	Interpreting the parameter	6
4	The Binomial Distribution	6
4.1	Random experiment	6
4.2	PMF formula	7
4.3	Why the binomial coefficient appears	7
4.4	Expected value and variance	7
4.5	Typical probability questions	8
4.6	When not to use the binomial model	8
5	The Hypergeometric Distribution	9
5.1	Sampling without replacement	9
5.2	Support	9
5.3	PMF formula	9
5.4	Expected value and variance	10
5.5	Binomial versus hypergeometric	10
6	The Geometric Distribution	11
6.1	Waiting for the first success	11
6.2	PMF formula	11
6.3	Cumulative and tail probabilities	11
6.4	Expected value and variance	12
6.5	Memoryless property	12
7	The Poisson Distribution	12

7.1	Counting events in a fixed interval	12
7.2	Meaning of the parameter	13
7.3	Changing the interval length	13
7.4	Poisson as an approximation to binomial	14
8	The Multinomial Distribution	14
8.1	More than two categories	14
8.2	PMF formula	15
8.3	Connection with the binomial distribution	15
8.4	Expected values and covariances	15
9	A Structural Comparison of the Main Models	16
10	Choosing a Distribution	16
11	Worked Conceptual Examples	17
11.1	Fixed trials with two outcomes	17
11.2	Sampling from a finite population	17
11.3	Waiting for first success	18
11.4	Counting arrivals in a time interval	18
11.5	Several categories	18
12	Common Misunderstandings	18
12.1	Confusing elementary outcomes with values of a random variable	18
12.2	Using binomial when trials are not independent	18
12.3	Forgetting the support	18
12.4	Misreading “at least” and “at most”	19
12.5	Treating expected value as a guaranteed value	19
13	Simulation and Empirical Interpretation	19
13.1	Theoretical probability versus simulated frequency	19
14	Summary of Formulas	20
15	Summary of Expectations and Variances	20
16	Final Conceptual Map	20

1 Purpose of These Notes

Probability theory begins with events, sample spaces, and rules for assigning probabilities. In many applications, however, we do not only ask whether an event happened. We often ask for a numerical summary of the outcome: the number of successful attempts, the number of defects, the waiting time until the first success, or the number of requests arriving in a fixed interval.

This is the role of a **random variable**. A probability distribution describes how the probability mass is assigned to the possible values of a random variable.

These notes are not a solution set. They are a general lecture on the mathematical structures behind selected discrete probability models:

- Bernoulli and binomial distributions,
- hypergeometric distribution,
- geometric distribution,
- Poisson distribution,
- multinomial distribution.

The goal is not to memorize several formulas independently. The goal is to understand what kind of random experiment each distribution represents, what assumptions justify the model, what the parameters mean, and what the random variable counts.

Main idea

A distribution is not just a formula. It is a mathematical model of a random mechanism. Before using a distribution, always identify:

1. the random experiment,
2. the elementary outcomes,
3. the random variable,
4. the support of the random variable,
5. the assumptions behind the model,
6. the meaning of the parameters.

2 From Experiments to Random Variables

2.1 Sample space and elementary outcomes

A **sample space** is the set of all elementary outcomes of a random experiment. It is denoted by Ω . An elementary outcome is a complete result of the experiment at the level of detail chosen by the model.

For example, if a device is tested five times and each test gives either success S or failure F , then one possible elementary outcome is

$$(S, F, S, S, F).$$

The sample space consists of all sequences of length five over the alphabet $\{S, F\}$:

$$\Omega = \{S, F\}^5.$$

This sample space has $2^5 = 32$ elementary outcomes.

2.2 Random variables

A **random variable** is a function from the sample space to a set of numbers:

$$X : \Omega \rightarrow \mathbb{R}.$$

It assigns a numerical value to each elementary outcome.

Random variable

A random variable is a numerical function of the outcome of a random experiment. It does not create randomness by itself; it extracts a number from an already random outcome.

In the previous testing example, define X as the number of successes. Then

$$X(S, F, S, S, F) = 3.$$

Many different elementary outcomes may lead to the same value of X . For instance,

$$(S, S, F, F, S), \quad (F, S, S, S, F), \quad (S, F, S, S, F)$$

all give $X = 3$.

2.3 Support of a random variable

The **support** of a discrete random variable is the set of values that the random variable can take with positive probability:

$$\text{supp}(X) = \{x \in \mathbb{R} : \mathbb{P}(X = x) > 0\}.$$

It is important not to confuse the sample space Ω with the support of X .

Sample space versus support

The sample space contains elementary outcomes. The support contains values of the random variable.

For five success/failure trials:

$$\Omega = \{S, F\}^5,$$

while the support of the number of successes is

$$\text{supp}(X) = \{0, 1, 2, 3, 4, 5\}.$$

These are different mathematical objects.

2.4 Probability distribution

For a discrete random variable X , the probability distribution is described by the probabilities

$$\mathbb{P}(X = x)$$

for all values x in the support.

A valid discrete probability distribution must satisfy:

$$\mathbb{P}(X = x) \geq 0$$

for every value x , and

$$\sum_{x \in \text{supp}(X)} \mathbb{P}(X = x) = 1.$$

Probability mass function

The function

$$p_X(x) = \mathbb{P}(X = x)$$

is called the **probability mass function**, or PMF, of the discrete random variable X .

2.5 Distribution as compression of a sample space

A distribution often compresses a large sample space into a smaller set of numerical values.

For example, three coin tosses have eight elementary outcomes:

$$HHH, HHT, HTH, HTT, THH, THT, TTH, TTT.$$

If X is the number of heads, then X has only four possible values:

$$0, 1, 2, 3.$$

The distribution records how probability is collected over these values.

Value of X	Outcomes	Probability for a fair coin
0	TTT	$1/8$
1	HTT, THT, TTH	$3/8$
2	HHT, HTH, THH	$3/8$
3	HHH	$1/8$

This table is already a probability distribution.

3 Bernoulli Trials and the Bernoulli Distribution

3.1 The two-outcome trial

A **Bernoulli trial** is a random experiment with two possible categories:

success or failure.

The word “success” does not necessarily mean something good. It means the event being counted.

If the probability of success is p , then the probability of failure is $1 - p$, where

$$0 \leq p \leq 1.$$

Bernoulli distribution

A random variable X has a Bernoulli distribution with parameter p , written

$$X \sim \text{Bernoulli}(p),$$

if

$$X = \begin{cases} 1, & \text{success occurs,} \\ 0, & \text{failure occurs,} \end{cases}$$

and

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

The support is

$$\text{supp}(X) = \{0, 1\}.$$

The expected value and variance are

$$\mathbb{E}[X] = p,$$

$$\text{Var}(X) = p(1 - p).$$

3.2 Interpreting the parameter

The parameter p is the probability that a single trial produces the outcome labeled as success. It is not the number of successes. It is not a percentage observed in a sample unless it has been estimated from data. In the model, p is an assumed probability.

Example: single alarm check

A monitoring system checks whether a server produces an alert in one minute. Let success mean “an alert occurs.” If the probability of an alert in a minute is $p = 0.03$, then

$$X \sim \text{Bernoulli}(0.03),$$

where $X = 1$ means alert and $X = 0$ means no alert.

4 The Binomial Distribution

4.1 Random experiment

The binomial distribution appears when we repeat independent Bernoulli trials a fixed number of times.

The model assumptions are:

1. There is a fixed number n of trials.
2. Each trial has two possible categories: success or failure.
3. The probability of success is the same in every trial and equals p .
4. The trials are independent.
5. The random variable counts the number of successes.

Binomial distribution

A random variable X has a binomial distribution with parameters n and p , written

$$X \sim \text{Bin}(n, p),$$

if X is the number of successes in n independent Bernoulli trials, each with success probability p .

The support is

$$\text{supp}(X) = \{0, 1, 2, \dots, n\}.$$

4.2 PMF formula

The probability of exactly k successes is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Each factor has a meaning:

Factor	Meaning
$\binom{n}{k}$	number of possible positions of the k successes among n trials
p^k	probability contribution of the k successes
$(1-p)^{n-k}$	probability contribution of the $n-k$ failures

4.3 Why the binomial coefficient appears

Consider a fixed sequence of n trials containing exactly k successes. For example, for $n = 6$ and $k = 2$, one such sequence is

$$SFFSFF.$$

Its probability is

$$p^2(1-p)^4.$$

But there are many sequences with exactly two successes. The number of such sequences is the number of ways to choose the positions of the successes:

$$\binom{6}{2}.$$

Therefore all sequences together have probability

$$\binom{6}{2} p^2 (1-p)^4.$$

4.4 Expected value and variance

If $X \sim \text{Bin}(n, p)$, then

$$\mathbb{E}[X] = np,$$

$$\text{Var}(X) = np(1-p).$$

The expected value np is the long-run average number of successes per block of n trials, not necessarily the value obtained in one concrete experiment.

Example: message classification

A filter marks each incoming message as suspicious with probability 0.12, independently from message to message. In a batch of 8 messages, let X be the number marked suspicious. Then

$$X \sim \text{Bin}(8, 0.12).$$

For exactly two suspicious messages,

$$\mathbb{P}(X = 2) = \binom{8}{2} (0.12)^2 (0.88)^6.$$

The expected number of suspicious messages is

$$\mathbb{E}[X] = 8 \cdot 0.12 = 0.96.$$

4.5 Typical probability questions

For $X \sim \text{Bin}(n, p)$, common questions include:

- exactly k successes:

$$\mathbb{P}(X = k),$$

- at most k successes:

$$\mathbb{P}(X \leq k) = \sum_{j=0}^k \binom{n}{j} p^j (1-p)^{n-j},$$

- at least k successes:

$$\mathbb{P}(X \geq k) = \sum_{j=k}^n \binom{n}{j} p^j (1-p)^{n-j},$$

- between a and b successes:

$$\mathbb{P}(a \leq X \leq b) = \sum_{j=a}^b \binom{n}{j} p^j (1-p)^{n-j}.$$

Often it is easier to use the complement. For example,

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - (1-p)^n.$$

4.6 When not to use the binomial model

The binomial model is not appropriate if:

- the number of trials is not fixed in advance,
- the trials are not independent,
- the probability of success changes from trial to trial,
- there are more than two categories and they are not reduced to success/failure,

- sampling is done without replacement from a small finite population.

Common mistake

The binomial distribution is not simply “any model with successes and failures.” It specifically requires a fixed number of independent trials with the same success probability.

5 The Hypergeometric Distribution

5.1 Sampling without replacement

The hypergeometric distribution appears when we sample from a finite population without replacement.

Suppose a population contains:

- N objects in total,
- K distinguished objects, called successes,
- $N - K$ non-distinguished objects, called failures.

We draw n objects without replacement. Let X be the number of successes in the sample.

Hypergeometric distribution

A random variable X has a hypergeometric distribution with parameters N, K, n , written

$$X \sim \text{Hypergeom}(N, K, n),$$

if X counts the number of successes in a sample of size n drawn without replacement from a finite population of size N , containing K successes.

5.2 Support

The value of X cannot be arbitrary. It must satisfy two restrictions:

- we cannot draw more successes than the population contains: $X \leq K$,
- we cannot draw more successes than the sample size: $X \leq n$,
- if too many failures are unavailable, some successes are forced.

Therefore

$$\text{supp}(X) = \{\max(0, n - (N - K)), \dots, \min(n, K)\}.$$

5.3 PMF formula

For $X \sim \text{Hypergeom}(N, K, n)$,

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

The denominator counts all possible samples of size n . The numerator counts samples containing exactly k successes and $n - k$ failures.

Factor	Meaning
$\binom{K}{k}$	choose k successes from K distinguished objects
$\binom{N-K}{n-k}$	choose $n - k$ failures from $N - K$ non-distinguished objects
$\binom{N}{n}$	choose any sample of size n from the full population

5.4 Expected value and variance

If $X \sim \text{Hypergeom}(N, K, n)$, then

$$\mathbb{E}[X] = n \frac{K}{N}.$$

The variance is

$$\text{Var}(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N - n}{N - 1}.$$

The final factor

$$\frac{N - n}{N - 1}$$

is called the finite population correction. It reflects the fact that drawing without replacement reduces variability compared with independent drawing.

Example: archive audit

An archive contains 40 documents, of which 9 are incomplete. An auditor randomly selects 6 documents without replacement. If X is the number of incomplete documents in the selected sample, then

$$X \sim \text{Hypergeom}(40, 9, 6).$$

The probability of exactly two incomplete documents is

$$\mathbb{P}(X = 2) = \frac{\binom{9}{2} \binom{31}{4}}{\binom{40}{6}}.$$

5.5 Binomial versus hypergeometric

The binomial and hypergeometric distributions both count successes. The difference is the sampling mechanism.

Feature	Binomial	Hypergeometric
Sampling mechanism	with replacement or independent trials	without replacement
Population	often conceptual or effectively infinite	finite and explicit
Success probability	constant p in every trial	changes after each draw
Independence	yes	no
Random variable	number of successes	number of successes

Model choice

Use the binomial model when trials are independent and the success probability stays fixed. Use the hypergeometric model when objects are drawn without replacement from a finite population and the composition changes after each draw.

6 The Geometric Distribution

6.1 Waiting for the first success

The geometric distribution describes the waiting time until the first success in independent Bernoulli trials.

The assumptions are:

1. Trials are repeated until the first success occurs.
2. Each trial has two outcomes: success or failure.
3. The probability of success is constant and equals p .
4. Trials are independent.
5. The random variable records the trial number on which the first success occurs.

Geometric distribution

A random variable X has a geometric distribution with parameter p , written

$$X \sim \text{Geom}(p),$$

if X is the number of the trial on which the first success occurs.

In these notes, the convention is

$$\text{supp}(X) = \{1, 2, 3, \dots\}.$$

Some books use a different convention where the random variable counts the number of failures before the first success. Always check the convention.

6.2 PMF formula

For $k = 1, 2, 3, \dots$,

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p.$$

This formula says:

- the first $k - 1$ trials must be failures,
- the k -th trial must be a success.

6.3 Cumulative and tail probabilities

The probability that the first success occurs no later than trial k is

$$\mathbb{P}(X \leq k) = 1 - (1 - p)^k.$$

The probability that the first success occurs after trial k is

$$\mathbb{P}(X > k) = (1 - p)^k.$$

This tail probability is often the simplest way to compute geometric probabilities.

6.4 Expected value and variance

If $X \sim \text{Geom}(p)$ with support $\{1, 2, 3, \dots\}$, then

$$\mathbb{E}[X] = \frac{1}{p},$$

$$\text{Var}(X) = \frac{1-p}{p^2}.$$

The expected waiting time decreases when p increases.

Example: network reconnection

A device attempts to reconnect to a network. Each attempt succeeds with probability 0.25, independently of previous attempts. Let X be the attempt number on which the first successful reconnection occurs. Then

$$X \sim \text{Geom}(0.25).$$

The probability that the first success is on the fourth attempt is

$$\mathbb{P}(X = 4) = (0.75)^3(0.25).$$

The probability that the device needs more than five attempts is

$$\mathbb{P}(X > 5) = (0.75)^5.$$

6.5 Memoryless property

The geometric distribution has an important property called **memorylessness**. For $m, n \geq 0$,

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n).$$

This means that, conditional on having waited m unsuccessful trials, the remaining waiting time has the same distribution as at the beginning.

Interpretation of memorylessness

Memorylessness does not mean that the past did not happen. It means that, under the model assumptions, past failures do not change the probability law of future trials.

7 The Poisson Distribution

7.1 Counting events in a fixed interval

The Poisson distribution models the number of events occurring in a fixed interval of time, space, area, length, or volume.

The typical assumptions are:

1. events occur one at a time,
2. events occur independently in disjoint intervals,
3. the average rate of occurrence is stable over the interval,

4. the probability of two or more events in a very small interval is negligible.

Poisson distribution

A random variable X has a Poisson distribution with parameter $\lambda > 0$, written

$$X \sim \text{Pois}(\lambda),$$

if X counts the number of events in a fixed interval and

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

The support is

$$\text{supp}(X) = \{0, 1, 2, \dots\}.$$

7.2 Meaning of the parameter

The parameter λ is the expected number of events in the interval:

$$\mathbb{E}[X] = \lambda.$$

For a Poisson distribution,

$$\text{Var}(X) = \lambda.$$

Thus the mean and variance are equal. This is a strong structural property of the model.

Example: sensor alerts

A sensor produces on average 2.4 alerts per day. If the number of alerts in one day is modeled as Poisson, then

$$X \sim \text{Pois}(2.4).$$

The probability of no alerts on a given day is

$$\mathbb{P}(X = 0) = e^{-2.4}.$$

The probability of at least one alert is

$$\mathbb{P}(X \geq 1) = 1 - e^{-2.4}.$$

7.3 Changing the interval length

If events occur at rate r per unit interval, then over an interval of length t the Poisson parameter is

$$\lambda = rt.$$

For example, if calls arrive at an average rate of 6 per hour, then in half an hour the expected number is

$$\lambda = 6 \cdot 0.5 = 3.$$

The parameter must correspond to the interval being analyzed.

Common mistake

Do not use a rate for one interval length while computing probabilities for another interval length. If the interval changes, λ must be rescaled.

7.4 Poisson as an approximation to binomial

If $X \sim \text{Bin}(n, p)$, where n is large, p is small, and $np = \lambda$ is moderate, then

$$X \approx \text{Pois}(\lambda).$$

This approximation is useful for rare events observed over many opportunities.

Poisson approximation

The Poisson approximation is natural when we count rare events among many independent opportunities. The parameter is usually

$$\lambda = np.$$

8 The Multinomial Distribution**8.1 More than two categories**

The multinomial distribution generalizes the binomial distribution from two categories to several categories.

Suppose each trial produces one of r possible categories:

$$1, 2, \dots, r.$$

The probabilities of the categories are

$$p_1, p_2, \dots, p_r,$$

where

$$p_i \geq 0, \quad p_1 + p_2 + \dots + p_r = 1.$$

We perform n independent trials. Let

$$X_i = \text{number of outcomes in category } i.$$

Then

$$X_1 + X_2 + \dots + X_r = n.$$

Multinomial distribution

The random vector

$$(X_1, \dots, X_r)$$

has a multinomial distribution with parameters n and $(p_1, \dots, p_r)$, written

$$(X_1, \dots, X_r) \sim \text{Multinomial}(n; p_1, \dots, p_r),$$

if it records category counts in n independent trials with category probabilities $p_1, \dots, p_r$.

8.2 PMF formula

For nonnegative integers $x_1, \dots, x_r$ satisfying

$$x_1 + x_2 + \dots + x_r = n,$$

the probability is

$$\mathbb{P}(X_1 = x_1, \dots, X_r = x_r) = \frac{n!}{x_1!x_2!\dots x_r!} p_1^{x_1} p_2^{x_2} \dots p_r^{x_r}.$$

The coefficient

$$\frac{n!}{x_1!x_2!\dots x_r!}$$

counts the number of sequences with exactly x_1 results of category 1, x_2 results of category 2, and so on.

8.3 Connection with the binomial distribution

If $r = 2$, the multinomial distribution reduces to the binomial distribution. In that case

$$(X_1, X_2) \sim \text{Multinomial}(n; p, 1 - p),$$

and knowing X_1 automatically determines $X_2 = n - X_1$. Therefore the distribution of X_1 is

$$X_1 \sim \text{Bin}(n, p).$$

8.4 Expected values and covariances

If

$$(X_1, \dots, X_r) \sim \text{Multinomial}(n; p_1, \dots, p_r),$$

then

$$\mathbb{E}[X_i] = np_i,$$

$$\text{Var}(X_i) = np_i(1 - p_i),$$

and for $i \neq j$,

$$\text{Cov}(X_i, X_j) = -np_i p_j.$$

The covariance is negative because, for a fixed total number of trials, more observations in one category leave fewer observations available for other categories.

Example: website traffic categories

A website classifies each visit as direct, search, social, or referral. Suppose the category probabilities are

$$(0.30, 0.45, 0.15, 0.10).$$

In $n = 20$ independent visits, let

$$(X_1, X_2, X_3, X_4)$$

be the counts in the four categories. Then

$$(X_1, X_2, X_3, X_4) \sim \text{Multinomial}(20; 0.30, 0.45, 0.15, 0.10).$$

The probability of counts $(5, 10, 3, 2)$ is

$$\frac{20!}{5!10!3!2!} (0.30)^5 (0.45)^{10} (0.15)^3 (0.10)^2.$$

9 A Structural Comparison of the Main Models

The distributions in these notes can be organized by the type of random mechanism they describe.

Distribution	Random mechanism	Random variable	Support
Bernoulli	one success/failure trial	success indicator	$\{0, 1\}$
Binomial	fixed number of independent success/failure trials	number of successes	$\{0, \dots, n\}$
Hypergeometric	sample without replacement from finite population	number of successes in sample	finite interval
Geometric	repeat independent trials until first success	trial number of first success	$\{1, 2, \dots\}$
Poisson	events in a fixed interval	number of events	$\{0, 1, 2, \dots\}$
Multinomial	fixed number of independent trials with several categories	vector of category counts	integer vectors summing to n

10 Choosing a Distribution

A practical way to choose a model is to ask what the random variable counts.

Distribution selection questions

1. Is there only one success/failure trial? Use Bernoulli.
2. Is there a fixed number of independent success/failure trials? Use binomial.
3. Is the sampling without replacement from a finite population? Use hypergeometric.
4. Is the experiment repeated until the first success? Use geometric.
5. Are we counting events in a fixed interval with a stable rate? Use Poisson.
6. Are there several categories in each of a fixed number of independent trials? Use multinomial.

Question	Natural model
One success/failure trial	Bernoulli
Fixed number of independent success/failure trials	Binomial
Sampling without replacement from a finite population	Hypergeometric
Repeated trials until the first success	Geometric
Counting random events in a fixed interval	Poisson
Fixed number of independent trials with several categories	Multinomial

The diagram is not a formal theorem. It is a guide to the modeling decision. Real situations may require judgment, approximation, or additional assumptions.

11 Worked Conceptual Examples

The examples below are included to illustrate how models are selected. They are not solutions to a task list.

11.1 Fixed trials with two outcomes

A quality scanner examines 12 independent items. For each item, define success as “the scanner reports a warning.” Suppose the warning probability is $p = 0.08$. Let X be the number of warnings.

The correct structure is:

- fixed number of trials: $n = 12$,
- two outcomes per trial: warning or no warning,
- constant success probability: $p = 0.08$,
- independent trials,
- random variable: number of warnings.

Thus

$$X \sim \text{Bin}(12, 0.08).$$

11.2 Sampling from a finite population

A library shelf has 50 books, of which 6 have damaged covers. A librarian selects 8 books without replacement. Let X be the number of books with damaged covers.

Here the success probability changes after each selection because the sample is taken without replacement. Therefore the structure is hypergeometric:

$$X \sim \text{Hypergeom}(50, 6, 8).$$

11.3 Waiting for first success

A login system sends one verification code at a time. Suppose each attempt is successfully received with probability 0.9, independently of other attempts. Let X be the attempt number on which the first code is successfully received.

The number of attempts is not fixed in advance. The experiment continues until first success. Therefore

$$X \sim \text{Geom}(0.9).$$

11.4 Counting arrivals in a time interval

Suppose a small service receives on average 4.5 support messages per hour, and messages are modeled as arriving randomly and independently over time. Let X be the number of messages in one hour.

The natural model is

$$X \sim \text{Pois}(4.5).$$

11.5 Several categories

A survey classifies each response into one of four categories: very satisfied, satisfied, neutral, dissatisfied. Suppose the category probabilities are

$$0.20, 0.45, 0.25, 0.10.$$

For 30 independent responses, let (X_1, X_2, X_3, X_4) be the category counts. Then

$$(X_1, X_2, X_3, X_4) \sim \text{Multinomial}(30; 0.20, 0.45, 0.25, 0.10).$$

12 Common Misunderstandings

12.1 Confusing elementary outcomes with values of a random variable

A sequence such as

$$SFSFFS$$

may be an elementary outcome. The number of successes in that sequence is a value of a random variable. The sequence and the number are not the same object.

12.2 Using binomial when trials are not independent

If objects are sampled without replacement from a small finite population, the trials are dependent. After each draw, the composition of the remaining population changes. A hypergeometric model is usually more appropriate.

12.3 Forgetting the support

Every probability formula is valid only on its support. For example:

- binomial: $k = 0, 1, \dots, n$,

- geometric: $k = 1, 2, 3, \dots$,
- Poisson: $k = 0, 1, 2, \dots$,
- hypergeometric: k must be feasible given N, K, n ,
- multinomial: all category counts must be nonnegative and sum to n .

12.4 Misreading “at least” and “at most”

The phrase “at least k ” means

$$X \geq k.$$

The phrase “at most k ” means

$$X \leq k.$$

For discrete distributions, these are sums of probabilities over several values, not just one value.

12.5 Treating expected value as a guaranteed value

The expected value is a long-run average, not a promise about a single experiment. If $X \sim \text{Pois}(2.4)$, then $\mathbb{E}[X] = 2.4$, even though X can only take integer values.

13 Simulation and Empirical Interpretation

Probability distributions are mathematical models. A simulation generates artificial outcomes according to such a model. By repeating the simulation many times, one can compare empirical frequencies with theoretical probabilities.

13.1 Theoretical probability versus simulated frequency

If $X \sim \text{Bin}(10, 0.3)$, then $\mathbb{P}(X = 3)$ is a theoretical number determined by the model. If we simulate the experiment 1000 times and observe how often $X = 3$, we obtain an empirical relative frequency.

The empirical frequency is random. It usually approaches the theoretical probability as the number of repeated simulations grows, but it does not have to be exactly equal to it.

Role of simulation

Simulation does not replace the mathematical model. It helps visualize the model, check intuition, and observe how random samples fluctuate around theoretical probabilities.

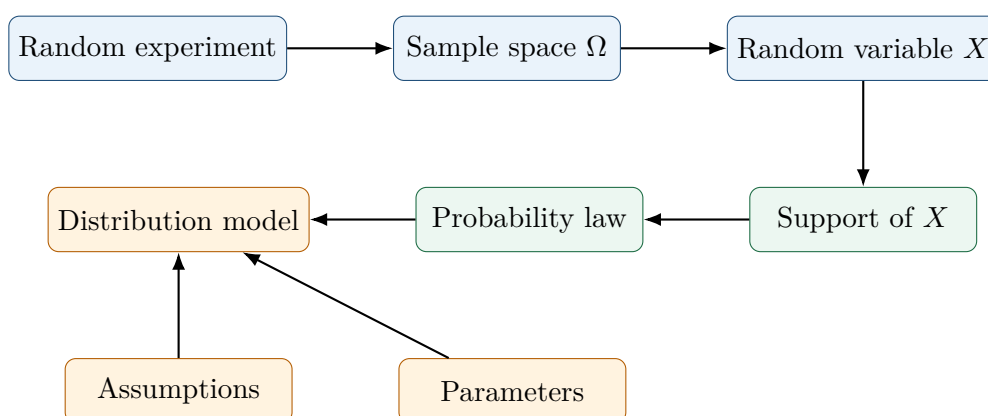
14 Summary of Formulas

Distribution	Probability formula	Parameters
Bernoulli	$\mathbb{P}(X = 1) = p, \mathbb{P}(X = 0) = 1 - p$	p
Binomial	$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$	n, p
Hypergeometric	$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$	N, K, n
Geometric	$\mathbb{P}(X = k) = (1 - p)^{k-1} p$	p
Poisson	$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$	λ
Multinomial	$\frac{n!}{x_1! \cdots x_r!} \prod_{i=1}^r p_i^{x_i}$, where $x_1 + \cdots + x_r = n$	$n; p_1, \dots, p_r$

15 Summary of Expectations and Variances

Distribution	Expected value	Variance
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$
Hypergeometric(N, K, n)	nK/N	$n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$
Geometric(p)	$1/p$	$(1 - p)/p^2$
Poisson(λ)	λ	λ
Multinomial component X_i	np_i	$np_i(1 - p_i)$

16 Final Conceptual Map



A probability distribution should always be read as part of this chain. The formula is only the final visible layer of the model. The mathematical meaning comes from the experiment, the random variable, the assumptions, and the interpretation of parameters.